Schur's Theorem

Proof by Mathematical Induction

1 Unitary Matrix

For an $n \times n$ complex matrix Q, if

$$\overline{Q}^T = Q^{-1},$$

then Q is called a **unitary matrix**.

If a unitary matrix is real, then

$$\overline{Q} = Q,$$

and hence

$$Q^T = Q^{-1}.$$

Thus, in the real case, a unitary matrix is an **orthogonal matrix**.

2 Schur's Theorem

Theorem 1 (Schur's Theorem)**.** *Every square matrix can be factored as*

$$\boxed{A = QTQ^{-1} = QT\overline{Q}^T,}$$

where T is upper triangular and Q is unitary.

If A has real eigenvalues, then Q and T can be chosen real. In this case,

$$Q^T = Q^{-1},$$

so Q is orthogonal.

3 Proof

We prove the theorem by mathematical induction.

1. Base Case

For $n = 1$,

$$a = 1 \cdot a \cdot 1^{-1}.$$

2. Induction Hypothesis

Assume the theorem holds for every $k \times k$ matrix, and let A be a $(k+1) \times (k+1)$ matrix.

Let λ_1 be an eigenvalue of A and q_1 a corresponding unit eigenvector:

$$Aq_1 = \lambda_1 q_1, \qquad \|q_1\| = 1.$$

Using Gram–Schmidt, extend q_1 to an orthonormal basis

$$q_1, q_2, \ldots, q_{k+1}.$$

Let

$$Q_1 = \begin{bmatrix} q_1 & q_2 & \cdots & q_{k+1} \end{bmatrix}.$$

Then Q_1 is unitary.

3. Compute $\overline{Q}_1^T A Q_1$

Write the multiplication explicitly:

$$\overline{Q}_1^T A Q_1 = \begin{bmatrix} \overline{q}_1^T \\ \overline{q}_2^T \\ \vdots \\ \overline{q}_{k+1}^T \end{bmatrix} \begin{bmatrix} Aq_1 & Aq_2 & \cdots & Aq_{k+1} \end{bmatrix}.$$

Since

$$Aq_1 = \lambda_1 q_1,$$

we get

$$\overline{Q}_1^T A Q_1 = \begin{bmatrix} \overline{q}_1^T \lambda_1 q_1 & \overline{q}_1^T A q_2 & \cdots & \overline{q}_1^T A q_{k+1} \\ \overline{q}_2^T \lambda_1 q_1 & \overline{q}_2^T A q_2 & \cdots & \overline{q}_2^T A q_{k+1} \\ \vdots & \vdots & \ddots & \vdots \\ \overline{q}_{k+1}^T \lambda_1 q_1 & \overline{q}_{k+1}^T A q_2 & \cdots & \overline{q}_{k+1}^T A q_{k+1} \end{bmatrix}.$$

Because the q_i are orthonormal,

$$\overline{q}_1^T q_1 = 1, \qquad \overline{q}_j^T q_1 = 0 \quad (j = 2, \ldots, k+1).$$

Hence the first column becomes

$$\begin{bmatrix} \lambda_1 \\ 0 \\ \vdots \\ 0 \end{bmatrix},$$

and therefore

$$\overline{Q}_1^T A Q_1 = \begin{bmatrix} \lambda_1 & * & * & \cdots & * \\ 0 & & & & \\ 0 & & A_2 & & \\ \vdots & & & & \\ 0 & & & & \end{bmatrix}.$$

Here A_2 is a $k \times k$ matrix.

4. Apply the Induction Hypothesis to A_2

By the induction hypothesis,

$$A_2 = Q_2 T_2 \overline{Q}_2^T,$$

where Q_2 is unitary and T_2 is upper triangular. Hence

$$\overline{Q}_2^T A_2 Q_2 = T_2.$$

Let

$$Q = Q_1 \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & & & & \\ 0 & & Q_2 & & \\ \vdots & & & & \\ 0 & & & & \end{bmatrix}.$$

Denote the second matrix by D, so that $Q = Q_1 D$.

5. Show that Q is Unitary

Since Q_2 is unitary,

$$Q_2 \overline{Q}_2^T = I_k,$$

and hence

$$D \overline{D}^T = I_{k+1}.$$

Also,

$$Q_1 \overline{Q}_1^T = I_{k+1}.$$

Therefore

$$Q \overline{Q}^T = Q_1 D \overline{D}^T \overline{Q}_1^T = I_{k+1}.$$

Thus Q is unitary.

6. Compute $\overline{Q}^T AQ$

We have

$$\overline{Q}^T AQ = \overline{D}^T \overline{Q}_1^T AQ_1 D.$$

Writing out the matrices,

$$\overline{Q}^T AQ = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & & & & \\ 0 & & \overline{Q}_2^T & & \\ \vdots & & & & \\ 0 & & & & \end{bmatrix} \begin{bmatrix} \lambda_1 & * & * & \cdots & * \\ 0 & & & & \\ 0 & & A_2 & & \\ \vdots & & & & \\ 0 & & & & \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & & & & \\ 0 & & Q_2 & & \\ \vdots & & & & \\ 0 & & & & \end{bmatrix}.$$

Therefore

$$\overline{Q}^T AQ = \begin{bmatrix} \lambda_1 & * & * & \cdots & * \\ 0 & & & & \\ 0 & & \overline{Q}_2^T A_2 Q_2 & & \\ \vdots & & & & \\ 0 & & & & \end{bmatrix}.$$

Since

$$\overline{Q}_2^T A_2 Q_2 = T_2,$$

we obtain

$$\boxed{\overline{Q}^T AQ = \begin{bmatrix} \lambda_1 & * & * & \cdots & * \\ 0 & & & & \\ 0 & & T_2 & & \\ \vdots & & & & \\ 0 & & & & \end{bmatrix} = T.}$$

Since T_2 is upper triangular, T is upper triangular. Thus

$$\overline{Q}^T AQ = T.$$

Because

$$\overline{Q}^T = Q^{-1},$$

we have

$$Q^{-1} AQ = T,$$

and therefore

$$\boxed{A = QTQ^{-1} = QT\overline{Q}^T.}$$

Hence Schur's theorem is proved.